

Prefill-Context Execution Eval

20-Task Report with Baseline Comparison

Model: Claude Sonnet 4.6 | 2026-05-15 | SkillsBench Dataset

1. Experiment Overview

This experiment measures how Claude Sonnet 4.6 performs software engineering tasks when procedural SKILL.md documents are injected into the system prompt (prefill context). Three conditions are compared:

- Prefill (main experiment): GT skills + noise skills in candidate pool
- Noskill baseline: No skills injected, model uses only its own knowledge
- GT-only baseline: Only ground-truth skills injected, no noise

Design Parameters

Tasks	20 selected (14 lightweight executed, 6 heavy-dep skipped)
Prefill Configs	168 executed (4 pool sizes x 3 noise modes x 14 tasks)
Baseline Configs	28 executed (noskill + gtonly x 14 tasks)
Pool Sizes (prefill)	5, 10, 20, 50 candidates
Noise Modes (prefill)	random, hard (cosine-similar), easy (dissimilar)
Execution	Docker containers, 4x parallel, Max Plan credentials

2. Overall Results - Baseline Comparison

Condition	Pass	Total	Rate	vs Noskill
Noskill (no skills)	6	14	42.9%	-
GT-only (perfect retrieval)	2	14	14.3%	-28.6pp
Prefill avg (GT + noise)	93	168	55.4%	+12.5pp

Key finding: Noskill baseline achieves 43% pass rate. Prefill (GT + noise) achieves 55%, a +12pp delta. GT-only achieves only 14%, significantly worse than both noskill and prefill. This suggests that injecting only GT skills without noise context may confuse the model or create over-reliance on potentially misaligned instructions.

3. Per-Task Baseline Comparison

Task	Noskill	GT-only	Prefill	Skill Effect
3d-scan-calc	PASS	FAIL	100%	neutral
earthquake-plate-calculation	FAIL	FAIL	100%	HELPS
energy-market-pricing	FAIL	FAIL	100%	HELPS
dialogue-parser	PASS	FAIL	92%	neutral
adaptive-cruise-control	PASS	FAIL	83%	neutral
dapt-intrusion-detection	FAIL	PASS	83%	HELPS
earthquake-phase-association	PASS	PASS	75%	neutral
econ-detrending-correlation	PASS	FAIL	75%	neutral
energy-ac-optimal-power-flow	FAIL	FAIL	33%	neutral
exceltable-in-ppt	FAIL	FAIL	33%	neutral
azure-bgp-oscillation-route-..	FAIL	FAIL	0%	neutral
citation-check	PASS	FAIL	0%	HURTS
civ6-adjacency-optimizer	FAIL	FAIL	0%	neutral
enterprise-information-searc..	FAIL	FAIL	0%	neutral

4. Skill Injection Effect Analysis

Skills HELP (noskill=FAIL, prefill $\geq$ 50%)

- dapt-intrusion-detection: FAIL->>83% (skills provide critical procedural knowledge)
- earthquake-plate-calculation: FAIL->>100% (skills provide critical procedural knowledge)
- energy-market-pricing: FAIL->>100% (skills provide critical procedural knowledge)

Skills HURT (noskill=PASS, prefill<50%)

- citation-check: PASS->>0% (skill injection degrades performance)

Skills NEUTRAL

Tasks where noskill and prefill agree (both pass or both fail). For these tasks, skill injection neither helps nor hurts.

Both pass: 3d-scan-calc, adaptive-cruise-control, dialogue-parser, earthquake-phase-association, econ-detrending-correlation

Both fail: azure-bgp-oscillation-route-leak, civ6-adjacency-optimizer, energy-ac-optimal-power-flow, enterprise-information-search, exceltable-in-ppt

GT-only Anomaly

GT-only baseline (14%) performs dramatically worse than both noskill (43%) and prefill (55%). Only 2/14 tasks pass with GT-only (dapt-intrusion-detection, earthquake-phase-association). Hypotheses:

- GT skills may contain instructions that conflict with the model's default approach
- Without noise context, the model over-indexes on a single skill document
- Some GT SKILL.md files may have quality issues (e.g., azure-bgp FORMAT_MISMATCH)
- The model may need diversity of context to reason effectively

5. Noise Mode Analysis

Condition	Pass/Total	Rate
random	30/56	53.6%
hard	33/56	58.9%
easy	30/56	53.6%
noskill (baseline)	6/14	42.9%
gtonly (baseline)	2/14	14.3%

All three prefill noise modes outperform GT-only but only marginally outperform noskill. Hard noise (58.9%) slightly leads, suggesting semantically similar context provides marginal benefit. The narrow spread (53.6-58.9%) indicates the model is robust to distractor type when sufficient candidates are available.

6. Pool Size Analysis

Pool Size	Pass/Total	Rate	Avg Prompt
0 (noskill)	6/14	42.9%	0K
5	21/42	50.0%	17.3K
10	25/42	59.5%	20.9K
20	25/42	59.5%	32.0K
50	22/42	52.4%	56.1K

Including the noskill baseline (pool=0) reveals that skill injection provides a modest improvement over no context. The sweet spot is at pool sizes 10-20 (59.5%) versus noskill at 42.9%. Very large pools (50) show slight degradation.

7. Noise x Pool Size Matrix

Noise \ Pool	sz=5	sz=10	sz=20	sz=50
random	43%	57%	57%	57%
hard	57%	64%	57%	57%
easy	50%	57%	64%	43%

8. Key Findings

1. Prefill vs Noskill: Skill injection improves pass rate from 43% to 55% (+12pp). Skills provide meaningful but modest value.
2. GT-only Paradox: Injecting ONLY ground-truth skills (14%) is WORSE than no skills (43%). This is the most surprising finding - pure GT context hurts more than noise+GT.
3. Task Difficulty Dominates: Pass rate varies 0-100% across tasks, dwarfing noise mode (5.3pp) and pool size (9.5pp) effects.
4. Skills Are Critical for Some Tasks: earthquake-plate-calculation and energy-market-pricing go from 0% (noskill) to 100% (prefill). These tasks require specific procedural knowledge the model lacks.
5. Skills Hurt citation-check: Goes from PASS (noskill) to 0% (all prefill configs). The injected skills likely override correct model behavior.
6. Noise Provides Useful Diversity: The prefill condition (GT+noise) outperforming GT-only suggests noise skills provide useful context diversity that helps the model reason more effectively, even when individual noise skills are irrelevant.

9. Limitations and Future Work

Limitations:

- Single seed (seed=0): No variance estimation. Multiple seeds needed for confidence intervals and statistical significance testing.
- Baselines are single-trial per task: noskill and gtonly have N=1 per task (vs N=12 per task for prefill). Higher baseline N would improve reliability.
- 6 tasks skipped due to heavy dependencies (poppler, nodejs, build-essential).
- citation-check and azure-bgp may have data quality issues affecting results.

Future Work:

- Multiple seeds (3-5) for variance estimation on all conditions
- Multiple baseline trials (N=5+) for noskill and gtonly
- Investigate GT-only paradox: is it specific GT skills or the single-document framing?
- Test with Claude Opus for model comparison
- Expand to full SkillsBench (88 tasks)

10. Appendix: Aggregation Tables

A. Prefill Results by Pool Size x Noise

Pool Size	Noise	N	Pass	Rate
5	easy	14	7	50.0%
5	hard	14	8	57.1%
5	random	14	6	42.9%
10	easy	14	8	57.1%
10	hard	14	9	64.3%
10	random	14	8	57.1%
20	easy	14	9	64.3%
20	hard	14	8	57.1%
20	random	14	8	57.1%
50	easy	14	6	42.9%
50	hard	14	8	57.1%
50	random	14	8	57.1%

B. Full Per-Task Comparison

Task	NS	GT	Prefill	Rate	Wall(s)
3d-scan-calc	P	F	12/12	100%	72
adaptive-cruise-control	P	F	10/12	83%	725
azure-bgp-oscillation-rou..	F	F	0/12	0%	34
citation-check	P	F	0/12	0%	1
civ6-adjacency-optimizer	F	F	0/12	0%	1
dapt-intrusion-detection	F	P	10/12	83%	102
dialogue-parser	P	F	11/12	92%	244
earthquake-phase-associat..	P	P	9/12	75%	673
earthquake-plate-calculat..	F	F	12/12	100%	117
econ-detrending-correlati..	P	F	9/12	75%	117
energy-ac-optimal-power-f..	F	F	4/12	33%	316
energy-market-pricing	F	F	12/12	100%	334
enterprise-information-se..	F	F	0/12	0%	284
exceltable-in-ppt	F	F	4/12	33%	202